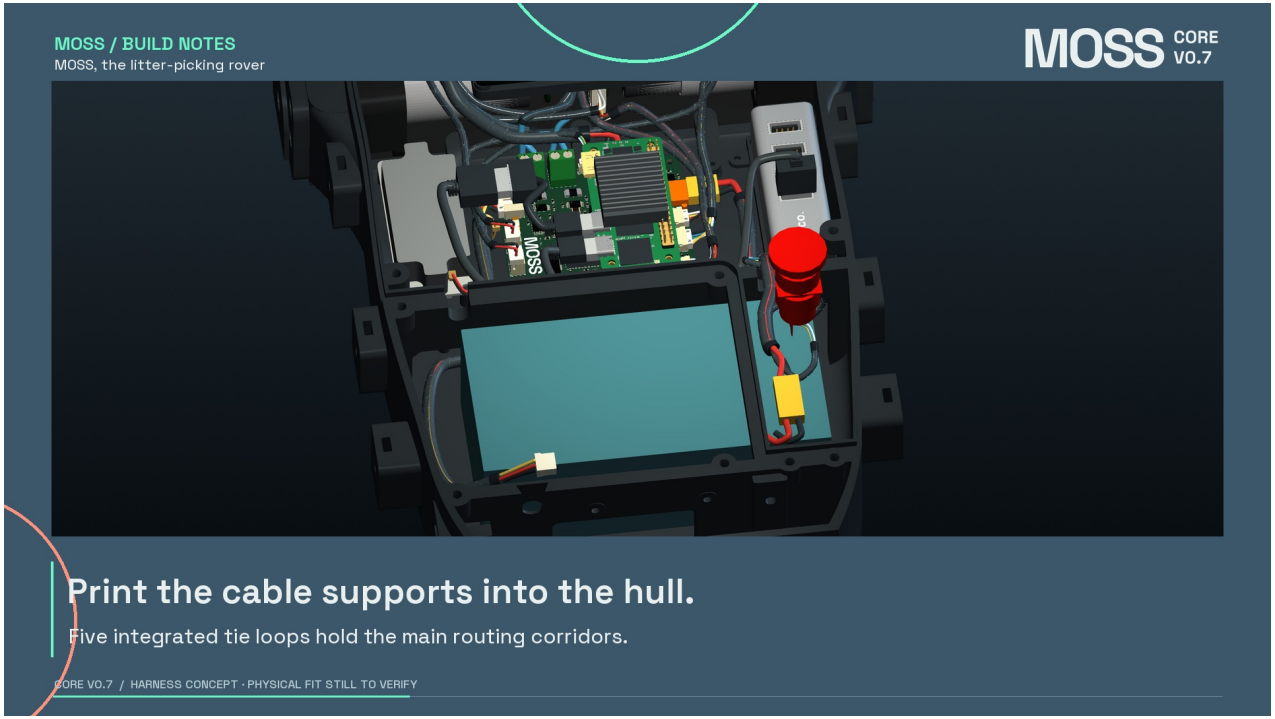
A ready-terminated wiring kit for MOSS

Prototype enquiry: two kits, plus separate prices at 10 / 50 / 100 kits

Please quote a labelled and electrically tested cable kit for MOSS, the litter-picking rover. The package now includes routed CAD, connector references and a wire schedule. Builders should not need to crimp or solder.



CAD study - the first kit must be fitted to the rover before batch lengths are frozen.

Base kit: H01-H08, including two-piece removable-lid STOP wiring. Each motor uses one shared six-wire plug.

Quote the documented STOP button with its wired lid tail separately. Reuse compatible arm, speaker and fan cables; adapters are optional.

Separate setup, tooling, test fixture, material sourcing, lead time and shipping to Thailand. No production or payment is authorized by this enquiry.

Cable lengths from the routed model

Free spans between connector wire exits; mm; prototype allowance is stated, not hidden

Connector wire-exit to wire-exit, measured along the relaxed cable. Each quote span is the longest CAD conductor plus 20 mm, rounded up to 5 mm. Manufacturing cut/strip/contact engagement allowances are added by the assembler, not included in this free span. Prototype target tolerance +/-5 mm. Physical routing to be checked on first article.

Route	Function / quantity	CAD longest wire	Quote free span	Wire
H01	Battery supply / 2 wires	174.6	195	16 AWG
H02	Cytron supply / 2 wires	115.9	140	16 AWG
H03	Cytron control / 5 wires	141.7	165	26 AWG / OD 0.76-1.00 mm
H04	Left encoder / 4 wires	126.2	150	26 AWG / OD 0.76-1.00 mm
H05	Right encoder / 4 wires	172.7	195	26 AWG / OD 0.76-1.00 mm
H06a	Left motor / shunt input / 1 wire	103.3	125	16 AWG
H06b	Left motor / shunt output / 1 wire	44.6	65	16 AWG main leg; stock motor tail
H06c	Left motor / direct return / 1 wire	126.6	150	16 AWG main leg; stock motor tail
H07a	Right motor / shunt input / 1 wire	68.9	90	16 AWG
H07b	Right motor / shunt output / 1 wire	119.7	140	16 AWG main leg; stock motor tail
H07c	Right motor / direct return / 1 wire	146.9	170	16 AWG main leg; stock motor tail
H08_chassis	STOP / chassis half / 4 wires	329.0	350	28 AWG / OD 0.50-0.80 mm
H08_lid	STOP / removable lid tail / 4 wires	131.1	155	28 AWG / OD 0.50-0.80 mm

H04+H06 and H05+H07 share one stock motor pigtail each. Lengths refer to complete routed legs including retained pigtail portions; do not add the pigtail twice.

Core connections and motor branches

Pin numbers follow the current circuit.json; the motor driver remains external

Core	Pins / nets	Destination
J10	1 GND / 2 BAT_PROTECTED	Battery - / fused battery +
J11	1 ACTUATOR_SW / 2 GND	Cytron POWER + / -
J12	1 CYTRON_L_LEAD / 2 MOTOR_L_LEAD	Cytron M1A -> J12.1; J12.2 -> left motor M1
J13	1 CYTRON_R_LEAD / 2 MOTOR_R_LEAD	Cytron M2A -> J13.1; J13.2 -> right motor M1
J2	1 MDD_PWM1 / 2 MDD_DIR1 / 3 MDD_PWM2 / 4 MDD_DIR2 / 5 GND	Match named Cytron control signals; retain its supplied five-way 2510 housing
J3	1 GND / 2 ENC_3V3 / 3 EL_A_EXT / 4 EL_B_EXT	Left encoder: GND / VCC / channel A / channel B
J4	1 GND / 2 ENC_3V3 / 3 ER_A_EXT / 4 ER_B_EXT	Right encoder: GND / VCC / channel A / channel B

Motor current paths: Cytron M1A -> J12.1 -> on-board shunt -> J12.2 -> left motor M1; M1B goes directly to left motor M2. Right side uses Cytron M2A/J13/right motor M1, with M2B direct to motor M2. Neither J12 nor J13 is a ground connector.

Motor source: Arduitronics RU00129, JGB37-520 12 V / 110 rpm, supplied six-wire pigtail [S1]. The product photographs show M1, GND, C2, C1, VCC, M2. Retain that plug, trace its wires to these labels and record cavity numbering before first-article termination. Do not infer polarity from animation colours or substitute PH/XH by appearance.

Quote factory-made, insulated transitions where the stock motor wire cannot be crimped directly into the Core contact. Record the stock gauge/insulation and the splice process. Assign the two encoder channels to A/B in the first-article wirelist; commissioning verifies direction. Encoder supply is switched 3.3 V, never motor battery voltage.

STOP: two separate loops and a removable lid

Four active wires; Core J8 is a six-position housing with cavities 3 and 4 empty

Core cavity	Signal	X08 circuit	Switch terminal	Colour
J8.1	ESTOP_CUT_SOURCE	1	C1	blue
J8.2	ESTOP_CUT_RETURN	2	NC1	white
J8.5	ESTOP_STATUS_SOURCE	3	C2	green
J8.6	ESTOP_STATUS_RAW	4	NC2	yellow

New-kit switch: Yijia YJ-LA16-22ZS/A, red, non-illuminated, 16 mm mounting hole and 30 mm head [S2]. C1-NC1 and C2-NC2 close when released and open when pressed. NO1/NO2 unused. Factory-solder the four tails to the 2.8 x 0.5 mm tabs, insulate each tab, and respect the manufacturer soldering limits. The older LA16-11 code specifies only one changeover contact.

X08: chassis 51021-0400 + 4 x 50058-8000; lid 51047-0400 + 4 x 50133-8000. Map manufacturer circuits 1-4 straight through. 28 AWG stranded copper, insulation OD 0.50-0.80 mm including tolerance, fits both PicoBlade and Core JST SH [S3-S7]. No transition splices are needed in H08.

The inline pair is polarized and friction-retained. Place it near the button, accessible after slightly lifting the cover. Support the cables with the modelled service loop; unplug by the housings. The current unmarked button can be reused only after its two independent NC pairs are identified by continuity.

Test, power disconnected	CUT loop: J8.1-2	STATUS loop: J8.5-6
Button released, X08 connected	Closed	Closed
Button pressed	Open	Open
X08 disconnected	Open	Open
Between the two circuits	Isolated	Isolated

Parts and options

Installed quantities per kit; no duplicate motor plug or separate encoder plug

Connection	Housing / assembly	Contacts / termination	Wire specification
J2	1 x GHR-05V-S	5 x SSHL-002T-P0.2	26 AWG; OD 0.76-1.00 mm
J3/J4	2 x GHR-04V-S	8 x SSHL-002T-P0.2	26 AWG; OD 0.76-1.00 mm
J8	1 x SHR-06V-S	4 x SSH-003T-P0.2-H	28 AWG; populate 1,2,5,6 only
X08 chassis	1 x 51021-0400	4 x 50058-8000	28 AWG; OD 0.50-0.80 mm
X08 lid	1 x 51047-0400	4 x 50133-8000	28 AWG; OD 0.50-0.80 mm
J10 extension	1 x XT30U-F; 1 x XT30U-M	Integrated solder cups	16 AWG; omit if existing lead reaches
J11-J13	Existing Phoenix 1715721 blocks	6 Core-side single-wire ferrules	16 AWG (1.31 mm2); one wire per cavity
Motor ends	2 x supplied RU00129 pigtails	Preserve original crimps	Factory transitions to main loom
Cytron control	Supplied 5-way 2510 housing	Retain factory lead or qualified retermination	Follow printed signal names, not Core pin order

Optional adapters: arm J5 uses 50-37-5033 + 3 x 08-70-1040, 22 AWG, OD 1.15-1.90 mm; speaker J7 uses PHR-2 + 2 x SPH-002T-P0.5S, 24 AWG, OD 0.80-1.50 mm; fans J14/J15 each use XHP-2 + 2 x SXH-001T-P0.6, 26 AWG, OD 0.90-1.90 mm. Reuse factory device leads where they mate and have the correct polarity; price any required adapter separately.

Optional pinouts: J5.1 DATA / .2 actuator power / .3 GND; J7.1 speaker + / .2 speaker - (neither is ground); J14/J15.1 fused 3S battery + / .2 GND.

Optional CAD-based free spans: O01 260 mm; O02 85 mm; O03_left 275 mm; O03_right 170 mm.

Standard cables, not custom-spliced USB: Core J20 -> Pi POWER, C-to-C 5 V/3 A (quote 150 mm); HB1 downstream -> Core J1, A-to-C data (quote 200 mm). Pi DATA connects through an OTG adapter to the HB1 captive USB-A cable. Its 800 mm total lead must be stored in the chassis; the visual scene shows only its routed portion.

First article and references

Supplier drawing approval follows the quotation; the CAD is not a wiring-test record

Provide a numbered mating-face drawing, wire specification, crimp tooling and finished-length tolerances. Label every branch at both ends; one conductor per screw-terminal cavity. Quote all factory splices explicitly.

Test every kit for continuity, shorts and polarity; record the two STOP loops in all four states on page 4. Fit one kit with the lid closed/open and inspect strain relief, fan clearance and USB cable storage before a larger batch.

Remaining sample checks: stock motor pigtail map/gauge/OD; actual unmarked STOP pairs if reused; device-lead mates and installed lengths. These cannot be established from catalogue photographs. The new documented STOP and PicoBlade references remove the corresponding procurement ambiguity.

Ref	Source
S1	Arduitronics RU00129: motor kit, photos and specification image
S2	Yijia LA16: contact diagram p1; 22ZS/A selection p3
S3	Molex PicoBlade mating drawing: 51047 + 51021, p2
S4	Molex female terminal: 50058-8000
S5	Molex male terminal: 50133-8000
S6	JST SH and GH connector specifications
S7	JST GH terminal and insulation range
S8	Cytron MDD10A official product/manual
S9	Phoenix 1715721 terminals
S10	AMASS XT30U
S11	JST PH: speaker termination
S12	JST XH: fan termination
S13	Molex arm housing drawing
S14	Molex arm contact drawing

Full source URLs are in SOURCES.md. Native Core PCB SHA256: cffb4a378502802a778fcee618310e6f3fa5c49ad201b0c372728368ae3520d1